

H2BENCH · DESIGN DECISION RECORD

H2Bench recommendations accepted

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DOCUMENT CONTROL

Document ID	HBN-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2bench		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 to O4 decided by Amish on 2026-10-02 (HBN-DEC-001)
0.2	2026-09-26	Amish Chadha	Budget approved by Amish (\$885)
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item in Table 1 is "Decided by Amish, 2026-09-25: go with recommendation"; items without a recommendation (Table 2) remained open; O1 to O4 were decided by Amish on 2026-10-02 (HBN-DEC-001).

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record lists every item in `docs/REVIEW.md` (TRL 2 and TRL 3 sessions) and in HBN-DDR-001 that carried a recommendation, now decided, and what changed in the repo. Items without a recommendation stay "Proposed, awaiting Amish". Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and `trl` and `trl_target` stay at 3.

■ Options considered

The options for each item are those listed in `docs/REVIEW.md` (sessions 2026-09-25, /populate and TRL 3), HBN-DDR-001 and HBN-CAL-001 v0.1.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Storage (DDR-001 item 1)	Rigid 2 L tank at up to 300 kPa gauge; gasbag with pump as the fallback if R10 fails	Wording only (HBN-PRC-001 v0.4, HBN-DDR-001 v0.2)
D2	Electrolysis type (item 2)	PEM with deionized water, not alkaline	Wording only
D3	Fuel cell source (item 3)	Generic 12 W class stack with a published polarization curve as a purchase condition; branded stack as the fallback	Wording only
D4	Hydrogen measurement (item 4)	Pressure, volume and temperature, not a mass flow meter	Wording only
D5	Source (item 5)	Bench power supply in constant-current mode; solar later	Wording only
D6	H2Guard cost scope (item 6)	Costed in its own project, shipped with the bench	Wording only (already in R14 scope)
D7	Tank minimum pressure (item 7)	Kept above about 20 kPa gauge; new tank purged by three pressure cycles	Wording only
D8	Canopy hood (item 11), proposed as the working design	Canopy hood with a single high point for sensing and extraction	Wording only (already in the model)
D9	Budget (item 9)	Recommendation (a): raise <code>budget_usd</code> to about \$850, bench supply included, H2Guard costed separately	<code>project.yaml</code> <code>budget_usd</code> \$450 to \$850; R14 restated as \$850 including the supply (HBN-REQ-001 v0.4); README, HBN-PRB-001 v0.4, HBN-PRC-001 v0.4, BOM notes and HBN-CAL-001 v0.2 updated. The kit is \$885, so R14 is still not met, by \$35
D10	Bench power supply in the kit (item 10)	In the kit, as D9 states	BOM line 4 note; R14 scope
D11	Relief valve setting (item 13)	325 kPa gauge (was 350)	<code>bom/bom.csv</code> line 10; HBN-CAL-001 v0.2: inventory at full relief lift and 15 °C 10.5 L to 9.9 L, R5 at risk to met; at the relief setting 9.6 L to 9.0 L; stored pressure energy at the relief setting 776 J to 710 J; room release 0.032 % to 0.030 % by volume. HBN-DWG-001 Rev P1 to P2 notes
D12	High-pressure supply cut (item 13)	Pressure switch opens at 310 kPa gauge and a relay breaks the electrolyzer supply; the relief valve is a backup only	<code>cad/src/model.py</code> : cut switch added to the manifold (item 10); STEP and STL re-exported; HBN-DWG-001 Rev P2; media regenerated. BOM line 10 \$45 to \$60 (switch), line 14 \$30 to \$35 (relay): total \$865 to \$885. Mass 25.2 to 25.3 kg. R10 restated from 300 kPa to 332 kPa (relief setting plus check valve); membrane load at the relief setting 357 to 332 kPa, 317 kPa at the cut. The cut is hardwired; no firmware rule is needed

#	Item	Decision	What changed in the repo
D13	Vent needle valve (item 13)	Set so venting 300 to 20 kPa gauge takes 3 min or more	Wording only (already in the design)
D14	Plug-in 30 mA RCD (item 13)	Keep as BOM line 18	Wording only (already in the BOM)
D15	H2Guard interface (item 14)	Settle the valve voltage, the output that breaks the 9 A supply line, the hood extraction rate and the first user with the H2Guard project	Listed under cross-repo actions in docs/REVIEW.md ; H2Guard not edited. R8 stays at risk

No pitch or problem rewording was recommended, so the pitch and problem lines in `project.yaml` and `README.md` are unchanged; the README cost line now states the \$850 budget.

Items still open

Table 2. Items open on 2026-09-25, with their later decisions.

#	Item	Status
01	Curriculum and age group for the first worksheets. No recommendation was made	Decided by Amish, 2026-10-02: post-16 vocational learners on hydrogen or renewable energy technician courses (HBN-DEC-001)
02	Class demonstration or group practical first. No recommendation was made	Decided by Amish, 2026-10-02: instructor-run class demonstration first; group practicals after a recorded run of attended fills (HBN-DEC-001)
03	Aluminium or stainless tank. No recommendation was made	Decided by Amish, 2026-10-02: certified aluminium cylinder, about 2 L, stamped 1 MPa or more (HBN-DEC-001)
04	Mass (R13, 25.3 kg against 25 kg): accept, carry the supply separately, or use a 10 mm deck (saves 0.77 kg). No recommendation was made	Superseded by HBN-DDR-003 A1; decided by Amish, 2026-10-02: bench supply beside the bench and a 10 mm deck (HBN-DEC-001)
05	Cost (R14, \$885 against \$850): where to find \$35. No recommendation was made	Decided by Amish, 2026-09-26: budget set to \$885 (see below)

Consequences

- Requirement status (HBN-CAL-001 v0.2): met 10 (was 9), not met 2 (R9 purity, R14 cost), at risk 3 (was 4: R8, R10, R13), not verifiable at TRL 3 1 (R16).
- Documents bumped: HBN-PRB-001 v0.4, HBN-PRC-001 v0.4, HBN-REQ-001 v0.4, HBN-CAL-001 v0.2, HBN-DDR-001 v0.2; drawing HBN-DWG-001 Rev P2.
- Cross-repo action: settle the interlock interface with H2Guard (D15). Not edited here.
- TRL 4 work (building the bench, supplier confirmation of back-pressure and purity, purchasing, lab tests, firmware beyond a sketch) remains on hold.

■ Budget approved, 2026-09-26

On 2026-09-26 Amish wrote, in chat: "i approve all the budget items."

- Budget set to \$885 to cover the priced BOM: decided by Amish, 2026-09-26. This settles O5. The 18-line BOM is \$885 with the bench supply (H2Guard excluded), so R14 moves from not met to met, with no margin.
- Requirement status (HBN-CAL-001 v0.3): met 11, not met 1 (R9 purity), at risk 3 (R8, R10, R13), not verifiable at TRL 3 1 (R16).
- Files changed: `project.yaml` (`budget_usd` 850 to 885); HBN-REQ-001 v0.5; HBN-CAL-001 v0.3, `docs/04-calcs/sizing.py` (hard-coded budget 850 to 885) and `results.csv` ; HBN-PRB-001 v0.5 and HBN-PRC-001 v0.5 (budget figure); `README.md` ; `bom/bom-notes.md` ; `docs/REVIEW.md` .